

# Ziyi Sun

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## EDUCATION

- **Ocean University of China (985 / Double First-Class)** Sep. 2023 – Jun. 2027 (Expected)  
B.Eng. in Mechanical Design, Manufacturing and Automation Qingdao, China
  - GPA: **3.7/4.0** Rank: **1/56** Average: 89.2
  - Selected courses: Advanced Mathematics (97), Linear Algebra (93.5), Probability & Mathematical Statistics (93), Complex Variables & Integral Transforms (94.5), Numerical Methods (96), Mechanical Vibrations (95), Modern Mechanical Design Theory and Methods (100), Underwater Robotics (97.2)

## RESEARCH EXPERIENCE

- **Roboparty Lab** Jul. 2026 – Oct. 2026  
Frontier Research and Open Source Intern (Mentor: Jagger) Beijing, China
  - Contribute to frontier research and open-source development for embodied intelligence and robot learning systems.

## PROJECTS

- **Variable-Length Multimodal Biomimetic Robotic Fish for Deep-Sea Inspection** Dec. 2025 – Dec. 2026  
Role: Project Leader | Tools: SolidWorks, multimodal vision & sensing OUC SRDP
  - Designed a biomimetic robotic fish with continuously adjustable body length (1.2–1.5 m) for deep-sea cage cruise and narrow-space inspection.
  - Fused school-fish visual tracking with water-quality sensing for joint monitoring and early warning of aquaculture risks.
- **Cross-Domain Wire-Driven Biomimetic Eel Robot** Dec. 2024 – Nov. 2025  
Role: Project Leader | Tools: vision tracking, Gaussian Splatting, damage detection Provincial Innovation Program
  - Developed an eel-like robot with cruising and active depth control for floating offshore wind dynamic-cable inspection.
  - Integrated visual tracking, 3D reconstruction, and damage detection for cable modeling, defect localization, and risk warning.
- **Streaming Evaluation Framework for Deep-Sea Mining** 2025 – 2026  
Role: Lead Contributor | Tools: CAD modeling, multi-AUV simulation & benchmarking Research Project
  - Built a unified simulation and benchmarking pipeline comparing centralized and distributed multi-AUV nodule collection strategies.
  - Integrated CAD-based robot models and reproducible evaluation protocols for scalable underwater robot learning.
- **High-Stability Duck-Type Wave Energy Converter with Pendulum PTO** Dec. 2024 – Nov. 2025  
Role: Core Member | Tools: AQWA, genetic algorithm, SolidWorks National Innovation Program
  - Improved capture efficiency and output stability via hydrodynamics modeling, shape optimization, and a combined PTO with nonlinear adaptive regulation.

## PUBLICATIONS & PATENTS

C=CONFERENCE, S=IN SUBMISSION, P=PATENT, SW=SOFTWARE

### [C.1] A Two-Stage Learning Framework for Autonomous Obstacle Avoidance of an Eel-Like Robot.

Ziyi Sun, Luwen Li, Changhong He, Shuo Jiang, Zhuoyan Wang, Haoyuan Cheng\*.

IEEE International Conference on Real-time Computing and Robotics (IEEE RCAR), 2026. **Best Paper Finalist.**

Proposed a two-stage framework combining multimodal perception, expert demonstration, PPO, AHTA-CPG, and domain randomization; achieved 70%–90% success over 60 real water-tank trials.

### [C.2] Design of Biomimetic Robotic Eel with Omnidirectional Wire-Driven Body and Controlled Central Pattern Generator.

Ziyi Sun, et al.

IEEE International Conference on Computer, Control and Robotics (ICCCR), 2025.

Presented the mechanical design and control framework of a biomimetic robotic eel with an omnidirectional quadruple wire-driven body and CPG-based locomotion control.

[S.1] **CHEROE: Every Humanoid Skill as a Traject.**  
**Ziyi Sun**, Jingwen Chen, Yuxi Wang, Xiuze Xia, Long Cheng, Zhaoxiang Zhang, Junyu Dong.  
Under review at AAAI 2027 (CCF-A).  
Unified SkillMotion representation and Tracker interface with boundary compatibility assessment, trajectory fusion, and bridging-action retrieval for reusable multi-source humanoid skills.

[S.2] **A Streaming Evaluation Framework for Deep-Sea Mining: Comparing Centralized and Distributed Nodule Collection.**  
**Ziyi Sun**, Changhong He.  
Under review at *Journal of Marine Science and Engineering (JMSE)*.  
Developed a unified simulation and benchmarking pipeline for scalable robot learning, integrating CAD-based robot models and reproducible comparisons of centralized vs. distributed multi-AUV nodule collection.

[P.1] **A High-Stability PTO System for Duck-Type Wave Energy Converters.**  
**Ziyi Sun**, et al.  
Invention Patent (First Student Inventor).

[SW.1] **Underwater Polarized Light Field Prediction Software for Biomimetic Navigation V1.0.**  
**Ziyi Sun**.  
Software Copyright (No. 2024SR0140235).

SKILLS

- **Robot Platform:** Unitree G1
- **Programming & ML:** Python, C/C++, MATLAB, PyTorch, Isaac Gym/Lab, MuJoCo
- **Mechanical Design:** SolidWorks, AutoCAD
- **Research Focus:** Embodied AI, robot learning, sim-to-real transfer, multimodal perception, underwater robotics
- **Languages:** Chinese (Native); English (CET-4: 591, CET-6: 482)

HONORS AND AWARDS

|                                                                                                   |             |
|---------------------------------------------------------------------------------------------------|-------------|
| • Shandong Provincial Government Scholarship                                                      | 2025        |
| • First-Class Comprehensive Scholarship, Ocean University of China                                | 2024 & 2025 |
| • Outstanding Student / Outstanding Student Cadre, Ocean University of China                      | 2024 & 2025 |
| • National First Prize, National 3D Digital Innovation Design Competition                         | 2025 & 2026 |
| • National Second Prize, Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM)      | 2025        |
| • Honorable Mention, Mathematical Contest in Modeling (MCM)                                       | 2025        |
| • National Third Prize, 18th National College Student Energy Conservation Competition             | 2024        |
| • National Gold Award, 3rd “Chuangyi Cup” National Undergraduate Extracurricular Academic Contest | 2024        |
| • National Second Prize, 14th Marine Vehicle Design and Production Competition                    | 2025        |
| • National First Prize, CRRC Cup National Undergraduate Renewable Energy Competition              | 2026        |
| • Science and Technology Innovation Pioneer Team, College of Engineering, OUC                     | 2024        |

LEADERSHIP EXPERIENCE

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|--------------------------------------------------------------------------------------------------------------------|----------------|
| • Class Representative / Class Monitor, Ocean University of China                                                  | 2023 – Present |
| • Team Leader, College of Engineering Debate Team, OUC                                                             | 2023 – Present |
| • Director of Event Organization, Student Association for the Study of Socialism with Chinese Characteristics, OUC | 2023 – Present |

VOLUNTEER EXPERIENCE

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|-------------------------------------------------------------------------------|------|
| • Volunteer, Qingdao Marathon                                                 | 2023 |
| • Member, National College Student Volunteer Outreach Team (Zunyi Conference) | 2024 |
| • Outstanding Individual in Summer Social Practice, Ocean University of China | 2024 |